54. MULTI-BRANCH EXPENSE MATRIX ANALYSIS

1. AIM

To write and execute a Python program that reads monthly expenses for multiple branches into a 2D matrix (list of lists) and calculates total and average expenses for each branch.

2. PROBLEM STATEMENT

Multi-branch organizations need to track branch-wise spending to analyze performance and manage operational budgets effectively. Write a Python program that inputs the number of branches and months, reads monthly expenses into a 2D matrix, and computes the total and average expenses for each branch.

3. OBJECTIVES

• Input total number of branches and total number of months.
• Construct a 2D matrix using nested loops to collect monthly branch expenses.
• Compute row-wise sum to calculate total expenses for each branch.
• Calculate and output average monthly expenses per branch.

4. ASSUMPTIONS & REQUIREMENTS

Assumptions: Number of branches and months are positive integers. Expense amounts are numerical values.
Requirements: Python 3.x Interpreter, PyCharm / VS Code / IDLE environment.

5. ALGORITHM

Step 1: Read number of branches and months.
Step 2: Initialize empty 2D matrix expenses = [].
Step 3: Loop i from 0 to branches-1: Create row = [], loop j from 0 to months-1: input amount and append to row, then append row to expenses.
Step 4: Loop i from 0 to branches-1: Compute total = sum(expenses[i]), average = total / months, and display.
Step 5: Stop program.

6. PROGRAM FLOWCHART

7. PYTHON PROGRAM (Multi-Branch Expense Matrix Analysis.py)

```
branches = int(input("Enter number of branches: "))
months = int(input("Enter number of months: "))

expenses = []

for i in range(branches):
    row = []
    for j in range(months):
        amount = float(input("Enter expense: "))
        row.append(amount)
    expenses.append(row)

for i in range(branches):
    total = sum(expenses[i])
    print("Branch", i + 1, "Total:", total)
    print("Average:", total / months)
```

8. EXPLANATION OF THE CODE

• branches, months: Stores dimension counts for the 2D matrix.
• expenses: List of lists representing the expense matrix.
• row.append(amount): Populates monthly expense records per branch.
• sum(expenses[i]): Computes cumulative branch spending.

9. TRACING TABLE

Step	Input	Expense Matrix	Total	Average
Branch 1, Month 1	**1.0**	**[[1.0]]**	**-**	**-**
Branch 2, Month 1	**2.0**	**[[1.0], [2.0]]**	**-**	**-**

Output Branch 1	-	[[1.0], [2.0]]	1.0	1.0
Output Branch 2	-	[[1.0], [2.0]]	2.0	2.0

10. TEST CASES

Test Case	Input	Expected Result	Actual Result
TC 1 (Normal)	Branches: 2, Months: 1 Expenses: 1, 2	Branch 1 Total: 1.0, Avg: 1.0 Branch 2 Total: 2.0, Avg: 2.0	Branch 1 Total: 1.0, Avg: 1.0 Branch 2 Total: 2.0, Avg: 2.0

11. EXPECTED OUTPUT

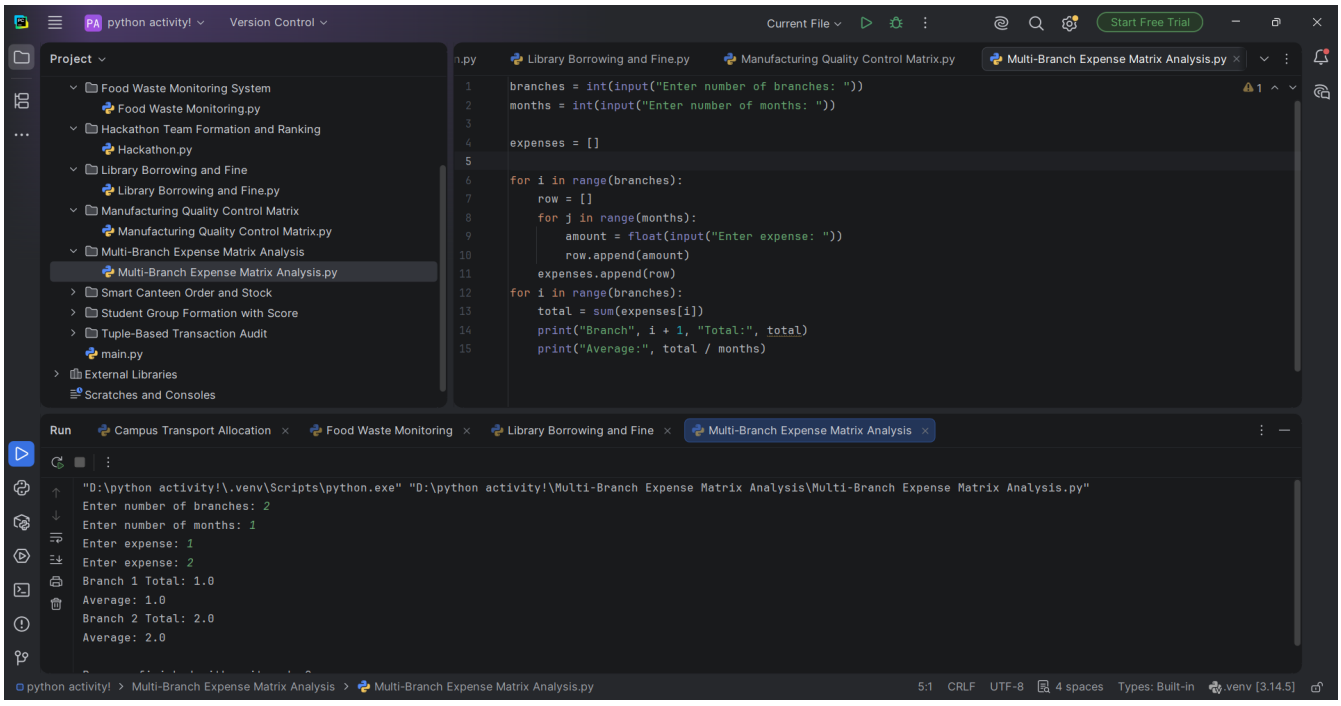
```
Enter number of branches: 2
Enter number of months: 1
Enter expense: 1
Enter expense: 2
Branch 1 Total: 1.0
Average: 1.0
Branch 2 Total: 2.0
Average: 2.0
```

12. ACTUAL OUTPUT

```
Enter number of branches: 2
Enter number of months: 1
Enter expense: 1
Enter expense: 2
Branch 1 Total: 1.0
Average: 1.0
Branch 2 Total: 2.0
Average: 2.0
```

13. PROGRAM EXECUTION SCREENSHOT

Actual Input and Output - PyCharm Run Screenshot



14. RESULT/CONCLUSION

The Python program for multi-branch expense matrix analysis was executed successfully. It accurately reads 2D branch data and calculates branch totals and averages.